

Incomplete (Missing) Data

- ❑ Data is not always available
 - ❑ E.g., many tuples have no recorded value for several attributes, such as customer income in sales data
- ❑ Missing data may be due to
 - ❑ Equipment malfunction
 - ❑ Inconsistent with other recorded data and thus deleted
 - ❑ Data were not entered due to misunderstanding
 - ❑ Certain data may not be considered important at the time of entry
 - ❑ Did not register history or changes of the data
- ❑ Missing data may need to be inferred

How to Handle Missing Data?

- ❑ Ignore the tuple: usually done when class label is missing (when doing classification)—not effective when the % of missing values per attribute varies considerably
- ❑ Fill in the missing value manually: tedious + infeasible?
- ❑ Fill in it automatically with
 - ❑ a global constant : e.g., “unknown”, a new class?!
 - ❑ the attribute mean
 - ❑ the attribute mean for all samples belonging to the same class: smarter
 - ❑ the most probable value: inference-based such as Bayesian formula or decision tree